In [1]:
```
import pandas as pd
```

In [3]:
```
df = pd.read_csv("Advanced_pandas_sales_dataset.csv")
```

In [4]:
```
df.head()
```

Out[4]:

	Order_ID	Order_Date	Customer	City	Region	Product	Category	Sales	Quantity	Discount	Profit
0	1001	2024-10-01	Priya	Hyderabad	East	Mouse	Accessories	1735	7	0.00	307.52
1	1002	2024-05-03	Sneha	Kolkata	East	Monitor	Electronics	807	2	0.15	49.89
2	1003	2024-01-26	Anita	Mumbai	West	Phone	Electronics	2328	1	0.20	66.47
3	1004	2024-06-06	Sara	Pune	East	Keyboard	Accessories	3933	3	0.20	-71.66
4	1005	2024-10-15	Rahul	Bangalore	South	Monitor	Electronics	3550	2	0.20	146.30

Q1 Dataset Exploration

Perform the following exploratory steps:

Display the first 10 rows

Display the dataset structure using info()

Find the number of rows and columns

Identify the data types of each column

In [5]:
```
print(df.head(10))
```

	Order_ID	Order_Date	Customer	City	Region	Product	Category	\
0	1001	2024-10-01	Priya	Hyderabad	East	Mouse	Accessories	
1	1002	2024-05-03	Sneha	Kolkata	East	Monitor	Electronics	
2	1003	2024-01-26	Anita	Mumbai	West	Phone	Electronics	
3	1004	2024-06-06	Sara	Pune	East	Keyboard	Accessories	
4	1005	2024-10-15	Rahul	Bangalore	South	Monitor	Electronics	
5	1006	2024-07-04	Pooja	Bangalore	West	Printer	Office	
6	1007	2024-06-24	Dev	Chennai	East	Monitor	Electronics	
7	1008	2024-03-18	Kavya	Kolkata	East	Phone	Electronics	
8	1009	2024-10-31	Kavya	Ahmedabad	East	Phone	Electronics	
9	1010	2024-12-25	Pooja	Ahmedabad	North	Phone	Electronics	

	Sales	Quantity	Discount	Profit
0	1735	7	0.00	307.52
1	807	2	0.15	49.89
2	2328	1	0.20	66.47
3	3933	3	0.20	-71.66
4	3550	2	0.20	146.30
5	3073	8	0.20	191.58
6	1170	5	0.20	22.83
7	4693	7	0.05	918.49
8	5071	6	0.10	707.55
9	2711	8	0.00	477.32

In [6]:
```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 230 entries, 0 to 229
Data columns (total 11 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Order_ID    230 non-null    int64
 1   Order_Date  230 non-null    object
 2   Customer    230 non-null    object
 3   City        230 non-null    object
 4   Region      230 non-null    object
 5   Product     230 non-null    object
 6   Category    230 non-null    object
 7   Sales       230 non-null    int64
 8   Quantity    230 non-null    int64
 9   Discount    230 non-null    float64
 10  Profit      230 non-null    float64
dtypes: float64(2), int64(3), object(6)
memory usage: 19.9+ KB
None
```

In [7]:
```
print(df.shape)
```

```
(230, 11)
```

In [8]:
```
print(df.dtypes)
```

```
Order_ID        int64
Order_Date     object
Customer       object
City           object
Region         object
Product        object
Category       object
Sales           int64
Quantity        int64
Discount      float64
Profit        float64
dtype: object
```

Q2 Sales Filtering Analysis

Filter the dataset to display:

Orders where Sales > 3500

Orders where Region = "East" and Sales > 2500

Explain what insights you observe.

In [9]:
```
df[df["Sales"] > 3500]
```

Out[9]:

	Order_ID	Order_Date	Customer	City	Region	Product	Category	Sales	Quantity	Discount	Profit
3	1004	2024-06-06	Sara	Pune	East	Keyboard	Accessories	3933	3	0.20	-71.66
4	1005	2024-10-15	Rahul	Bangalore	South	Monitor	Electronics	3550	2	0.20	146.30
7	1008	2024-03-18	Kavya	Kolkata	East	Phone	Electronics	4693	7	0.05	918.49
8	1009	2024-10-31	Kavya	Ahmedabad	East	Phone	Electronics	5071	6	0.10	707.55
10	1011	2024-06-30	Neha	Delhi	South	Printer	Office	5977	6	0.00	1578.32
...	...	...	...	...	...	...	...	...	...	...	...
219	1220	2024-03-22	Meera	Bangalore	West	Phone	Electronics	5986	7	0.05	986.70
222	1223	2024-03-04	Meera	Ahmedabad	South	Phone	Electronics	4356	6	0.10	416.87
223	1224	2024-01-07	Dev	Bangalore	East	Tablet	Electronics	4115	4	0.05	870.03
225	1226	2024-06-14	Anita	Ahmedabad	East	Laptop	Electronics	5647	2	0.15	662.00
229	1230	2024-04-13	Meera	Kolkata	North	Monitor	Electronics	5087	8	0.10	416.72

104 rows × 11 columns

In [10]:
```
df[(df["Region"] == "East") & (df["Sales"] > 2500)]
```

Out[10]:

	Order_ID	Order_Date	Customer	City	Region	Product	Category	Sales	Quantity	Discount	Profit
3	1004	2024-06-06	Sara	Pune	East	Keyboard	Accessories	3933	3	0.20	-71.66
7	1008	2024-03-18	Kavya	Kolkata	East	Phone	Electronics	4693	7	0.05	918.49
8	1009	2024-10-31	Kavya	Ahmedabad	East	Phone	Electronics	5071	6	0.10	707.55
13	1014	2024-03-18	Anita	Bangalore	East	Mouse	Accessories	3616	4	0.05	463.93
21	1022	2024-01-14	Nitin	Chennai	East	Monitor	Electronics	4827	6	0.05	912.34
27	1028	2024-07-21	Dev	Kolkata	East	Tablet	Electronics	4054	6	0.00	1013.83
33	1034	2024-12-15	Nitin	Ahmedabad	East	Tablet	Electronics	4378	2	0.20	-104.30
34	1035	2024-02-20	Nitin	Ahmedabad	East	Laptop	Electronics	2535	4	0.10	188.15
41	1042	2024-07-15	Rohan	Chennai	East	Laptop	Electronics	3268	8	0.15	295.15
43	1044	2024-09-20	Kavya	Hyderabad	East	Printer	Office	2618	7	0.05	454.64
52	1053	2024-05-24	Anita	Chennai	East	Phone	Electronics	5949	7	0.20	220.44
55	1056	2024-01-03	Sneha	Chennai	East	Headphones	Accessories	2784	4	0.05	474.60
58	1059	2024-12-30	Anita	Delhi	East	Tablet	Electronics	4790	1	0.20	238.53
60	1061	2024-02-05	Nitin	Delhi	East	Monitor	Electronics	4508	5	0.00	972.35
70	1071	2024-05-28	Vikas	Hyderabad	East	Mouse	Accessories	3762	2	0.05	719.68
71	1072	2024-05-20	Rahul	Chennai	East	Printer	Office	3696	2	0.10	600.75
73	1074	2024-10-10	Sara	Bangalore	East	Mouse	Accessories	4004	1	0.15	445.81
74	1075	2024-11-25	Pooja	Ahmedabad	East	Laptop	Electronics	3865	8	0.20	174.96
76	1077	2024-03-02	Neha	Pune	East	Monitor	Electronics	2964	8	0.20	109.33
81	1082	2024-08-16	Isha	Chennai	East	Phone	Electronics	2720	4	0.15	163.13
87	1088	2024-07-29	Pooja	Mumbai	East	Headphones	Accessories	4811	4	0.20	-25.47
92	1093	2024-01-13	Karan	Kolkata	East	Laptop	Electronics	5372	3	0.15	136.64

	Order_ID	Order_Date	Customer	City	Region	Product	Category	Sales	Quantity	Discount	Profit
96	1097	2024-06-07	Isha	Delhi	East	Phone	Electronics	5096	4	0.15	283.66
98	1099	2024-07-26	Rahul	Kolkata	East	Laptop	Electronics	5674	7	0.05	1141.51
108	1109	2024-10-23	Raj	Hyderabad	East	Tablet	Electronics	2509	4	0.05	371.67
109	1110	2024-11-28	Priya	Ahmedabad	East	Printer	Office	2561	4	0.20	57.96
112	1113	2024-11-01	Meera	Bangalore	East	Tablet	Electronics	4179	5	0.00	754.64
113	1114	2024-05-10	Rahul	Bangalore	East	Mouse	Accessories	3285	3	0.00	625.46
116	1117	2024-12-07	Pooja	Kolkata	East	Tablet	Electronics	3758	5	0.15	446.54
128	1129	2024-06-09	Nitin	Delhi	East	Laptop	Electronics	5713	3	0.00	1498.02
139	1140	2024-05-04	Karan	Ahmedabad	East	Monitor	Electronics	3500	8	0.05	467.16
141	1142	2024-01-01	Rahul	Mumbai	East	Headphones	Accessories	5430	8	0.05	741.25
146	1147	2024-05-31	Raj	Delhi	East	Keyboard	Accessories	5709	7	0.20	383.58
152	1153	2024-02-27	Rahul	Mumbai	East	Phone	Electronics	3644	2	0.05	512.25
156	1157	2024-05-27	Neha	Kolkata	East	Laptop	Electronics	4906	4	0.00	1115.89
171	1172	2024-09-22	Neha	Hyderabad	East	Headphones	Accessories	4228	1	0.20	236.15
179	1180	2024-12-13	Neha	Ahmedabad	East	Monitor	Electronics	2690	7	0.10	452.70
182	1183	2024-12-01	Pooja	Chennai	East	Laptop	Electronics	5884	1	0.05	1266.10
185	1186	2024-03-12	Kavya	Ahmedabad	East	Phone	Electronics	5047	2	0.10	564.78
187	1188	2024-07-10	Isha	Kolkata	East	Mouse	Accessories	4333	8	0.10	667.53
193	1194	2024-07-23	Pooja	Delhi	East	Laptop	Electronics	3767	7	0.20	103.23
197	1198	2024-10-16	Rahul	Ahmedabad	East	Tablet	Electronics	3955	1	0.10	284.92
202	1203	2024-05-10	Rahul	Mumbai	East	Phone	Electronics	2901	8	0.15	252.25
207	1208	2024-04-29	Dev	Chennai	East	Monitor	Electronics	3585	7	0.15	287.44

	Order_ID	Order_Date	Customer	City	Region	Product	Category	Sales	Quantity	Discount	Profit
216	1217	2024-08-16	Karan	Chennai	East	Monitor	Electronics	2643	5	0.15	78.53
223	1224	2024-01-07	Dev	Bangalore	East	Tablet	Electronics	4115	4	0.05	870.03
225	1226	2024-06-14	Anita	Ahmedabad	East	Laptop	Electronics	5647	2	0.15	662.00
228	1229	2024-09-04	Priya	Pune	East	Tablet	Electronics	2635	8	0.00	531.76

Insights:

Sales > 3500 represent high-value orders. East region has several orders with Sales > 2500, showing strong sales performance. These orders contribute significantly to overall revenue.

Q3 Customer Purchase Analysis

Identify the top 5 customers based on total sales.

In [11]:
```
customer_sales = df.groupby('Customer')['Sales'].sum()
```

In [12]:
```
customer_sales = customer_sales.sort_values(ascending=False)
```

In [13]:
```
top_5_customers = customer_sales.head(5)
```

In [14]:
```
print("Top 5 Customers Based on Total Sales:")
print(top_5_customers)
```

```
Top 5 Customers Based on Total Sales:
Customer
Nitin    59820
Raj      58416
Rahul    52291
Dev      48500
Isha     46818
Name: Sales, dtype: int64
```

Q4 Regional Sales Performance

Calculate the total sales generated by each region.

Then identify:

Which region generated the highest revenue?

In [15]:
```
region_sales = df.groupby('Region')['Sales'].sum()
```

In [16]:
```
print("Total Sales by Region:")
print(region_sales)
```

```
Total Sales by Region:
Region
East     232675
North    158210
South    183108
West     170227
Name: Sales, dtype: int64
```

In [17]:
```
print(region_sales.idxmax())
```

```
East
```

Q5 Product Performance Analysis

Calculate the total quantity sold for each product.

Which product appears to be the most popular?

In [18]:
```
product_quantity = df.groupby('Product')['Quantity'].sum()
```

In [19]:
```
print("Total Quantity Sold by Product:")
print(product_quantity)
```

```
Total Quantity Sold by Product:
Product
Headphones    115
Keyboard      110
Laptop        140
Monitor       156
Mouse         122
Phone         151
Printer       108
Tablet        150
Name: Quantity, dtype: int64
```

In [20]:
```
print(product_quantity.idxmax())
```

```
Monitor
```

Q6 Multi-Level Grouping

Group the dataset by: Region and Category

Calculate the total sales for each combination.

What insights can you derive from this analysis?

In [21]:
```
region_category_sales = df.groupby(['Region', 'Category'])['Sales'].sum()
```

In [22]:
```
print("Total Sales by Region and Category:")
print(region_category_sales)
```

```
Total Sales by Region and Category:
Region  Category
East    Accessories     57646
        Electronics    160907
        Office          14122
North   Accessories     62052
        Electronics     70023
        Office          26135
South   Accessories     80914
        Electronics     86690
        Office          15504
West    Accessories     64811
        Electronics     84128
        Office          21288
Name: Sales, dtype: int64
```

Insights:

East - Electronics generated the highest sales among all Region–Category combinations. South - Electronics and West - Electronics also showed strong sales performance. East region contributed the most revenue, especially through the Electronics category. Office category recorded comparatively lower sales across most regions. This analysis helps identify Electronics in the East region as the strongest sales segment and a key area for business focus.

Q7 Pivot Table Creation

Create a pivot table showing:

Rows → Region

Columns → Product

Values → Total Sales

Which region sells the most laptops?

In [23]:
```
pivot_table = pd.pivot_table(
    df,
    values='Sales',
```

```
    index='Region',
    columns='Product',
    aggfunc='sum'
)
```

In [24]:
```
print("Pivot Table:")
print(pivot_table)
```

```
Pivot Table:
Product  Headphones  Keyboard  Laptop  Monitor  Mouse  Phone  Printer  Tablet
Region
East          23899     10845   55568    31299  22902  39667    14122   34373
North         15450     20538    4540    14922  26064  30535    26135   20026
South         24111     23480   23029    19167  33323  18218    15504   26276
West          17159     38499   10826    18178   9153  27850    21288   27274
```

In [25]:
```
print(pivot_table['Laptop'].idxmax())
```

```
East
```

Insights:

East region sells the most Laptops, with total laptop sales of 55,568. South is the second-highest laptop-selling region with 23,029 in sales. North has the lowest laptop sales at 4,540. Overall, the East region shows the strongest performance across several products, especially Laptops.

In [ ]:
```
Q8 Profitability Analysis

Calculate the average profit for each product category.

Which category is most profitable?
```

In [26]:
```
category_profit = df.groupby('Category')['Profit'].mean()
```

In [27]:
```
print("Average Profit by Category:")
print(category_profit)
```

```
Average Profit by Category:
Category
Accessories    410.603086
Electronics    396.462114
Office         471.938462
Name: Profit, dtype: float64
```

In [28]:
```
print(category_profit.idxmax())
```

```
Office
```

Q9 Discount Impact Study

Analyze the relationship between discount and profit.

Tasks:

Calculate the average profit for each discount level

Identify whether higher discounts lead to lower profit

In [29]:
```
discount_profit = df.groupby('Discount')['Profit'].mean()
```

In [30]:
```
print("Average Profit by Discount Level:")
print(discount_profit)
```

```
Average Profit by Discount Level:
Discount
0.00    751.210588
0.05    531.368039
0.10    346.460513
0.15    237.191400
0.20     90.026923
Name: Profit, dtype: float64
```

Insights:

Profit is highest at 0% discount. As the discount increases, average profit generally decreases. Higher discounts reduce profit margins and can sometimes lead to very low or negative profits. Therefore, higher discounts tend to result in lower profit.

Q10 End-to-End Data Wrangling Task

Perform the following workflow:

Filter the dataset where Sales > 3000

Group data by Region and Product

Calculate total sales

Reshape the result into a pivot table

Explain the business insights obtained from this analysis.

In [31]:
```
filtered_data = df[df['Sales'] > 3000]
```

In [32]:
```
grouped_data = filtered_data.groupby(['Region', 'Product'])['Sales'].sum()
```

In [33]:
```
pivot_table = grouped_data.unstack()
```

In [34]:
```
print("Pivot Table:")
print(pivot_table)
```

```
Pivot Table:
Product  Headphones  Keyboard  Laptop  Monitor  Mouse  Phone  Printer  Tablet
Region
East          14469      9642   44096    16420  19000  29500     3696   29229
North          5326     15653    4540     5087  22209  24670    17185    3392
South         10026     19249   17950    14921  26928  18218    14143   19613
West          13898     33380    5922    14354   3511  15653    12648   17185
```

Insights:

The pivot table highlights which products generate the highest sales in each region for high-value orders (Sales > 3000). East region contributes the highest sales for most products. Laptops and Mobiles are among the strongest-performing products across regions. This analysis helps identify the best-selling products in each region and supports region-specific sales strategies.